Here is a detailed expansion of the concept, including specific tool proposals (technology stack) and implementation recommendations for each of the four AI Agents mentioned. The entire framework is based on my ballistic metaphor: instead of chasing the criminal and blocking domains, I examine the weapon and the infrastructure.

1. AI OSINT Crawler
Expansion:
This agent cannot be a simple script checking HTTP statuses. It must act as a virtual detective that understands the business context. Its task is to detect the semantic gap—the discrepancy between what the customer sees (e.g., a shoe store) and what the banking system sees (e.g., a construction company).

Workflow:

- Retrieves the URL and the tax/registry ID (e.g., NIP/REGON or foreign equivalent) provided during onboarding or transaction.
- Scrapes the website, categorizing the inventory (e.g., electronics).
- Queries government registry APIs (CEIDG/KRS/VIES), retrieving the company's NACE (PKD) business activity codes.
- Compares these two datasets. If the registered activity is 'social care' and the site is selling 'iPhone 15s', it generates an alert.

Recommended Technology Stack and Tools:

- **Browser Engine:** Puppeteer or Playwright in stealth mode to render Single Page Applications (SPAs) that a standard cURL request would miss.
- **NLP & Classification:** OpenAI API (e.g., GPT-4o-mini for speed) or a local BERT model trained on e-commerce categories to understand what the store is selling based on text and images.
- **Data Sources:** Integration with APIs like Transparent Data or direct access to the National Court Register (KRS/REGON) APIs.

2. Linguistic & Technical Auditor (Reverse Engineer)
Expansion:
Fraudsters operate globally, utilizing ready-made "fraud kits." They often forget to change default settings. This agent searches for the 'digital footprint' left behind in a hurry.

What exactly it analyzes:

- **Linguistic Artifacts:** Does a site on a .pl domain display error messages in Russian, Chinese, or Dutch (remnants of a previous campaign)?

- **Code Comments:** Are there leftover developer comments in the source code or variable names specific to known phishing templates?

- **DOM Structure:** Is the HTML structure 100% identical to a thousand other stores, despite different color schemes (known as HTML Structure Hashing)?

Recommended Technology Stack and Tools:

- **Code Analysis:** YARA Rules – the industry standard for creating rules to detect patterns in text/binary files.

- **Technology Fingerprinting:** Wappalyzer API or BuiltWith API to detect the engine powering the store (e.g., is it a WordPress site with a vulnerable plugin, or a fraudster's custom script?).

- **Hashing:** SSDEEP (Fuzzy Hashing) algorithm, which allows determining that two HTML files are 99% identical, even if the fraudster changed the logo and footer.

3. Network Linkage Engine (Connection Mapping)
Expansion:
This is the 'brain' of the operation. Fraudsters rarely build new infrastructure from scratch for every store. They usually attach new domains to old servers, use the same Google Analytics accounts, or the same DNS servers.

Workflow:

- The agent extracts 'fingerprints' from previously blocked stores (e.g., Google Tag Manager ID: GTM-XXXX).

- It searches the web or commercial databases for other sites using the same ID.

- It builds a relationship graph: if Store A (fraud) and Store B (new) share the same Pixel ID, Store B is preemptively blocked.

Recommended Technology Stack and Tools:

- **Graph Database:** Neo4j – ideal for storing relationships (Store -> IP -> Analytics ID <- Another Store).

- **Investigative Visualization:** Maltego – a tool for analysts to visualize connections.

- **Infrastructure Data:** RiskIQ (PassiveTotal), SecurityTrails API, or Censys – to check DNS history and IP address associations.

- **Tracker Analysis:** BuiltWith Relationships – shows all websites using the same analytics code.

4. Anomaly Detection in Payouts (Flow Guardian)
Expansion:
This agent looks at the bottom of the sales funnel. Fraudsters often use the 'Fan-In' technique: multiple different stores (domains) direct funds to a single bank account (or a group of associated 'mule' accounts).

Key Signals:

- **Multi-domain-to-One-IBAN:** 50 different stores (with varying goods) send money to the same account.

- **Velocity Check:** A new account that, within 24 hours, begins accepting hundreds of transactions of similar value (characteristic of selling non-existent goods on sale).

- **Geolocation Mismatch:** The store claims to operate in Poland, the banking login IP is from Asia, and the domains are registered in the Caribbean islands.

Recommended Technology Stack and Tools:

- **Analytics Platform:** Splunk or the ELK Stack (Elasticsearch, Logstash, Kibana) for real-time transaction log aggregation.

- **Stream Processing:** Apache Kafka + Apache Flink – for streaming analysis of transactions "on the fly."

- **AI Algorithms:** Isolation Forest models or Autoencoders (Scikit-learn or PyTorch libraries) – excellent for detecting anomalies in numerical data where rules are undefined (searching for the 'unknown unknown').

Strategic Recommendation: Integration into a 'Cyber-Loop'
Merely deploying tools is not enough. For the system to operate effectively, I recommend an Active Feedback Loop architecture.

- **Shadow Mode (Silent Phase):** Agents operate for 3 months without blocking, learning patterns from the bank's historical fraud data.

- **Human-in-the-Loop:** Agents do not block automatically (unless confidence is > 99%). They generate reports for SOC analysts.

- **Automated Enrichment:** If Agent 1 detects a suspicious store, it automatically tasks Agent 3 with checking its network connections and Agent 2 with deep code analysis.

- **Global Blacklist (Shared Intelligence):** Generated 'ballistic signatures' (e.g., a unique string in the fraudster's JavaScript code) should be immediately distributed to all banking systems.

  Conclusion:
  Deploying this architecture shifts the bank's role from a passive payment processor (which refunds money post-factum via chargeback procedures) into an active shield that neutralizes the threat before the funds ever leave the ecosystem.